



Technical Service Bulletin

Technical Service Bulletin: tsb100212

Released Date: 02-Jul-2003

Revised Cylinder Head Gasket

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This Technical Service Bulletin introduces the following:

- Cylinder Head Gasket Supersession With Latest Part Numbers and Changes
- QSK19 and K-19/38/50 Cylinder Head Gasket Identification
- Service Usage Guidelines for QSK19-profile and K-profile Cylinder Head Gaskets
- QSK19-profile Cylinder Head Gasket Improvement
- Engine Serial Number (ESN) Firsts and Implementation Dates of the Improved QSK19-profile Gasket
- Introduction of 0.51 mm [0.02 in] Oversize QSK19-profile Cylinder Head Gasket

This Technical Service Bulletin supersedes 99T2-1. Replace topic 99T2-1 with this topic.

The table below shows the new part numbers of each gasket and gives a brief summary of the changes made.

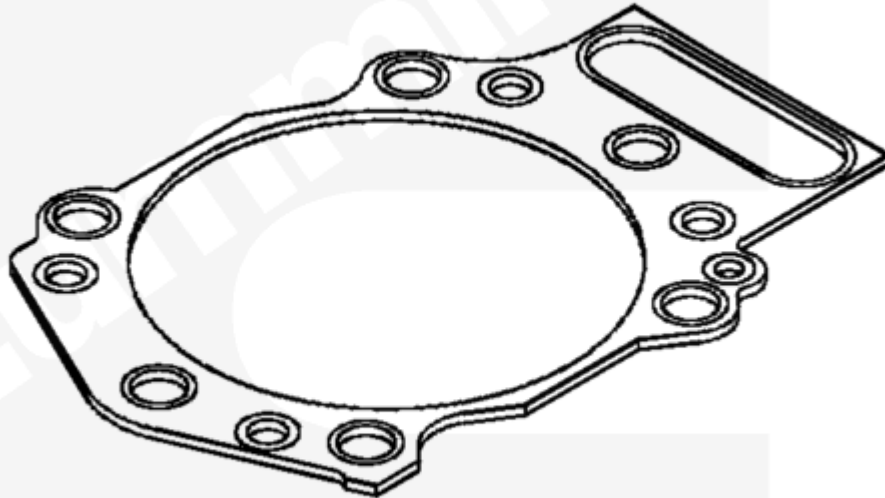
Gasket Type	Thickness	Old Part Number	New Part Number	Gasket Change
QSK19-profile	standard	3166289	3634664	Steel bridge added
QSK19-profile	0.51 mm [0.02 in] oversize	N/A	3634788	New oversize gasket
K-profile	standard	3166288	3166288	No change
K-profile	0.51 mm [0.02 in] oversize	3166287	3166287	No change

Two types of cylinder head gaskets are compatible with QSK19 and K19/38/50 engines:

The QSK19-profile gasket, Part Number 3634664 (3634788 oversize), is pictured in Figure 1 below. The coolant and oil seals are edge moulded to this gasket.

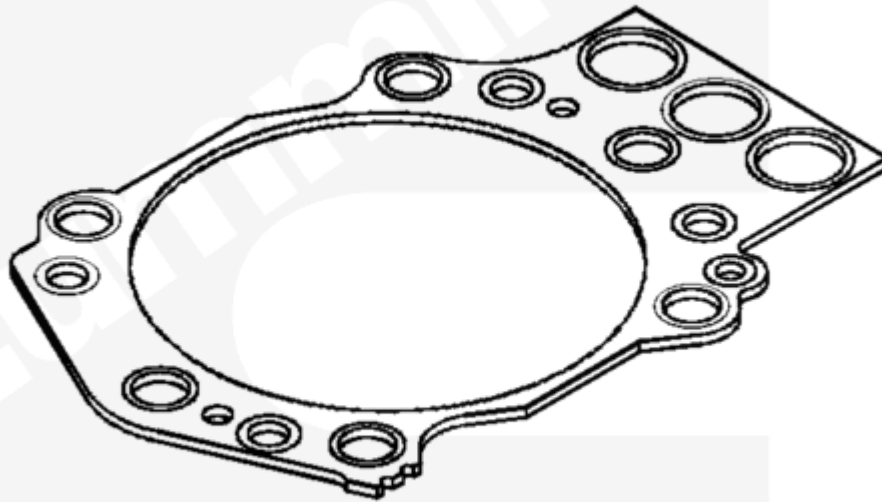
The K-profile gasket, Part Number 3166288 (3166287 oversize), is pictured in Figure 2 below. The K-profile gasket uses glued-in grommets to seal the coolant and oil passages.

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Figure 1, QSK19-profile



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Figure 2, K-profile

The QSK19-profile gasket and the K-profile gasket are geometrically different:

- The QSK19-profile gasket (Figure 1) has a single oblong hole surrounding all three push tubes. The overall length on the push tube end of the QSK19-profile gasket is approximately 2 mm [.08 in] longer than the K-profile gasket, causing the edge of the gasket to protrude beyond the edge of the cylinder head.
- The K-profile gasket has individual holes around each push tube.

⚠ CAUTION ⚠

If care is not taken when installing the intake manifold and manifold gaskets, there is potential for interference between the cylinder head gaskets and the intake manifold gaskets that can deform both gaskets, and possibly cause leaks. Make sure the intake manifold gaskets are not extending down over the head gasket, by applying an upward force on the intake manifold or aftercooler as the capscrews are tightened.

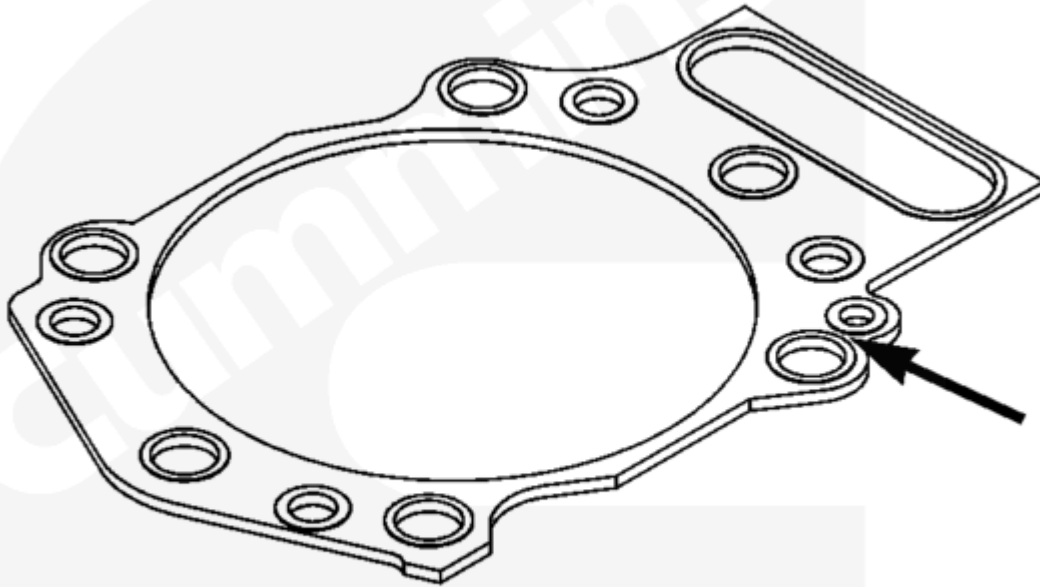
All QSK19 and K19/38/50 engines are manufactured with the QSK19-profile cylinder head gasket. The K-profile head gasket is available for service.

Use the QSK19-profile gasket if the mating surface of the block and head are refreshed during repair, or if the cylinder head gasket is being replaced.

Use the K-profile gasket when the mating surface of the cylinder block and head are **not** refreshed during repair, or if a K-profile gasket is being replaced.

Field experience has found, in some cases, the K-profile gasket provides better sealing characteristics than the QSK19-profile gasket on worn K19/38/50 cylinder blocks. A K-profile gasket can **not** be used on a QSK19 cylinder block.

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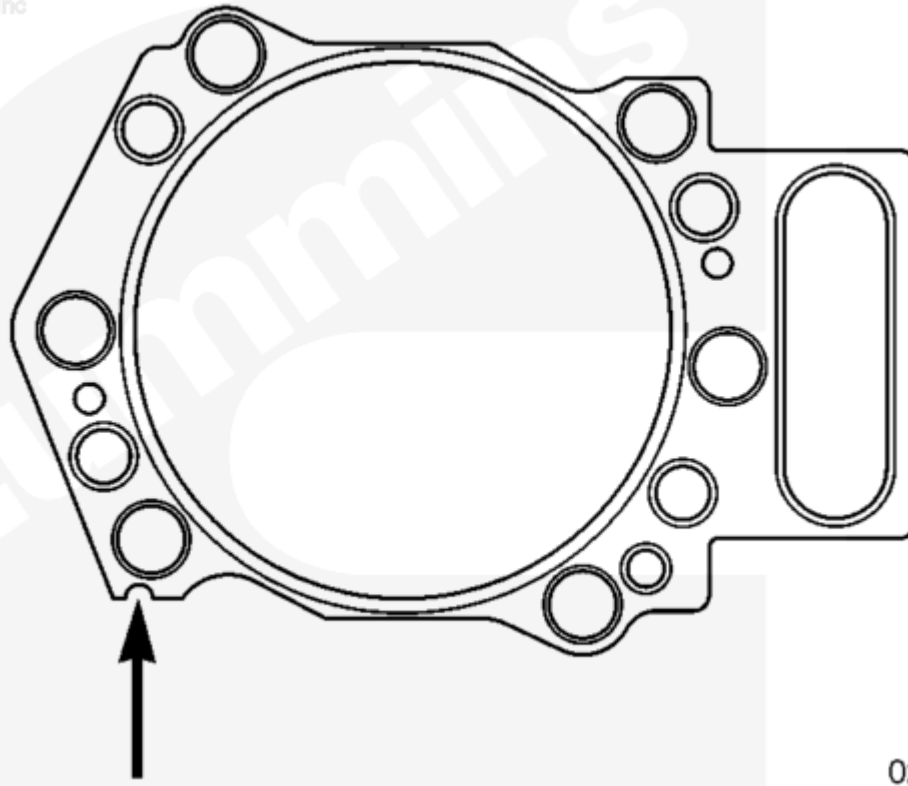
Figure 3, QSK19-profile Steel Bridge Enhancement

Note : The gasket referred to in this topic as the QSK19-profile gasket is common to all QSK19 and K19/38/50 engines.

As a product improvement, a steel bridge has been added to the gasket carrier between one of the head bolt drillings and the high pressure oil passage. The location of this durability enhancement is shown by the arrow in Figure 3 above.

The new QSK19-profile cylinder head gasket, Part Number 3634664 was implemented in production at the following dates and ESN firsts.

Engine	Date	ESN
K19/QSK19	12-12-2002	37206052
K38/50	01-27-2003	33153376



02400125

Figure 4, Oversize Gasket Notch

An oversize QSK19-profile cylinder head gasket, Part Number 3634788 has been developed and is now ready for service.

This gasket is 0.51 mm [0.02 in] thicker than the standard gasket and allows for greater machining of the cylinder block deck for rebuild or overhaul. A notch in the steel carrier to denotes the gasket as a service oversized gasket, shown by the arrow in Figure 4 above.